MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 5 Solution

---

1. Prove that the series $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges by showing that the sequence (s_k) of partial sums is increasing and bounded above.

   Solution.

   Let $s_k = \frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{k^2}$ be the k-th partial sum and it is increasing in k. Moreover,

   $$\begin{aligned} s_k = \frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{k^2} &\le 1+\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k-1)\cdot k} \\ &= 1+\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{k-1}-\frac{1}{k}\right) \\ &\le 2-\frac{1}{k}\le 2. \end{aligned}$$

   s_k is bounded above by 2, it converges by MCT. The series converges. ■

2. Fing a formula for the k-th partial sum s_k of the series

   $$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}+\sqrt{n+1}}$$

   and determine whether or not the series converges.

   Solution.

   $$\begin{aligned} s_k &= \frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\cdots+\frac{1}{\sqrt{k}+\sqrt{k+1}} \\ &= (\sqrt{2}-\sqrt{1})+(\sqrt{3}-\sqrt{2})+\cdots+(\sqrt{k+1}-\sqrt{k}) = \sqrt{k+1}-1 \end{aligned}$$

   The partial sum s_k diverges and hence the series also diverges.

   ■

3. Interpret the repeating decimal 0.424242... as a geometric series and express it as a rational number.

Solution.

$$0.424242... = 0.42 + 0.0042 + 0.000042 + \cdots = \sum_{n=0}^{\infty}(0.42)(0.01)^n = \frac{0.42}{1-0.01} = \frac{42}{99} = \frac{14}{33}$$

■

4. Let a_n and b_n be two sequence of positive real numbers and $\lim_{n\to\infty} \frac{a_n}{b_n} = L \geq 0$.

   (a) Prove that if $L > 0$, then $\sum_{n=1}^{\infty} a_n$ converges $\iff$ $\sum_{n=1}^{\infty} b_n$ converges.

   (b) Prove that if $L = 0$, then $\sum_{n=1}^{\infty} b_n$ converges $\implies$ $\sum_{n=1}^{\infty} a_n$ converges.

   (c) Determine if the series $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+n-1}{27n^5-n^4+9n^3-2n+6}$ converges.

   Solution.

   (a) Suppose $L > 0$, Take $\varepsilon = \frac{L}{2} > 0$, then there exists $N \in \mathbb{N}$ s.t.

   $$n > N \implies \left|\frac{a_n}{b_n} - L\right| < \frac{L}{2} \iff -\frac{L}{2} < \frac{a_n}{b_n} - L < \frac{L}{2} \iff \frac{L}{2}b_n < a_n < \frac{3L}{2}b_n$$

   If $\sum a_n$ converges, $\sum \frac{L}{2}b_n$ converges by the comparison test, and hence $\sum b_n$ also converges. Conversely, If $\sum b_n$ converges, $\sum \frac{3L}{2}b_n$ converges and hence $\sum a_n$ also converges.

   (b) Suppose $L = 0$, Take $\varepsilon = 1 > 0$, then there exists $N \in \mathbb{N}$ s.t.

   $$n > N \implies \left|\frac{a_n}{b_n}\right| < 1 \iff 0 < a_n < b_n$$

   If $\sum b_n$ converges, $\sum a_n$ converges by the comparison test.

   (c) Let $a_n = \dfrac{n^2+n-1}{27n^5-n^4+9n^3-2n+6} > 0$ and $b_n = \dfrac{1}{n^3} > 0$. We have,

   $$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{1}{27} > 0$$

   $\sum b_n$ converges (p-series with $p = 3$) and hence $\sum a_n$ also converges.

■

5. Prove the the series

$$\frac{1^2}{0!} + \frac{2^2}{1!} + \frac{3^2}{2!} + \cdots$$

converges and find its sum.

Solution.

Let $a_n = \frac{(n+1)^2}{n!} > 0$, $\lim\limits_{n\to\infty} \dfrac{a_{n+1}}{a_n} = \lim\limits_{n\to\infty} \dfrac{(n+2)^2 n!}{(n+1)^2(n+1)!} = \lim\limits_{n\to\infty} \dfrac{1}{n+1}\left(\dfrac{n+2}{n+1}\right)^2 = 0 < 1$

$\sum a_n$ converges by the ratio test.

$$\begin{aligned}
\frac{(n+2)^2}{n!} = \frac{n^2}{n!} + \frac{2n}{n!} + \frac{1}{n!} &= \frac{n}{(n-1)!} + \frac{2}{(n-1)!} + \frac{1}{n!} \qquad n \geq 2 \\
&= \frac{n-1+1}{(n-1)!} + \frac{2}{(n-1)!} + \frac{1}{n!} \\
&= \frac{1}{(n-2)!} + \frac{3}{(n-1)!} + \frac{1}{n!} \\
\implies \sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{(n+1)^2}{n!} &= \frac{1^2}{0!} + \frac{2^2}{1!} + \sum_{n=2}^{\infty} \frac{(n+1)^2}{n!} \\
&= 1 + 4 + \sum_{n=2}^{\infty} \left(\frac{1}{(n-2)!} + \frac{3}{(n-1)!} + \frac{1}{n!} \right) \\
&= 1 + 4 + [e + 3(e-1) + (e-2)] = 5e
\end{aligned}$$

■

6. Determine if each of the following series converges absolutely, conditionally, or diverges.

   (a) $\displaystyle\sum_1^{\infty} \frac{(-1)^{n+1}}{n+\sqrt{n}}$

   (b) $\displaystyle\sum_1^{\infty} \frac{(-1)^{n+1} n}{n+\sqrt{n}}$

Solution.

(a) $\displaystyle\sum_1^{\infty} \left|\frac{(-1)^{n+1}}{n+\sqrt{n}}\right| = \sum_1^{\infty} \frac{1}{n+\sqrt{n}}$, $\frac{1}{n+\sqrt{n}} \geq \frac{1}{2n}$, it diverges by comparison with the harmonic series. It does not converge absolutely.

$0 \leq \frac{1}{n+\sqrt{n}} \to 0$ is decreasing to zero. The series converges by the alternating series test.

(b) $\left|\dfrac{(-1)^{n+1} n}{n+\sqrt{n}}\right| \geq \dfrac{n}{n+\sqrt{n}} \geq \dfrac{1}{2}$. The series dicerges.

■

— *End* —